



**Tertiary Infotech Academy Pte Ltd**

UEN: 201200696W

# **LEARNER GUIDE**

For

## **Certified Lean Six Sigma White Belt (CLSSWB) Training**

Course Code: C142

Conducted by

**Tertiary Infotech Academy Pte Ltd**

UEN: 201200696W

**Version 3**

## DOCUMENT VERSION CONTROL RECORD

| Version Number | Effective Date of Release | Summary of Included Changes                                                                                                                                                                                                                  | Author         |
|----------------|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| 1              | 1 July 2026               | Initial release — CLSSWB 1-day lesson plan for the Lean Six Sigma awareness course.                                                                                                                                                          | Dr. Alfred Ang |
| 2              | 19 July 2026              | Rebuilt from the single-source content module: DMAIC roadmap end to end within one 8-hour training day, exactly one hands-on lab per DMAIC phase (5 labs) on a single continuous scenario, content simplified to White Belt awareness depth. | Dr. Alfred Ang |
| 3              | 21 July 2026              | Released under course code C142 as a commercial one-day hands-on course. Schedule restructured so every DMAIC lab gains additional hands-on time and the day closes with a full course recap and Q&A.                                        | Dr. Alfred Ang |

## TABLE OF CONTENTS

|                                                                       |           |
|-----------------------------------------------------------------------|-----------|
| <b>Introduction.....</b>                                              | <b>4</b>  |
| <b>Course Learning Outcomes.....</b>                                  | <b>4</b>  |
| <b>Before You Start — Preparation.....</b>                            | <b>4</b>  |
| <b>Topic 01 — Lean Six Sigma Foundations (20%).....</b>               | <b>5</b>  |
| <b>Topic 02 — Define — Understand the Problem (20%).....</b>          | <b>5</b>  |
| Lab 1 — Define — Customer Requirements and the Problem Statement..... | 6         |
| <b>Topic 03 — Measure — See What Is Really Happening (20%).....</b>   | <b>6</b>  |
| Lab 2 — Measure — Map the Process and Spot the Waste.....             | 7         |
| <b>Topic 04 — Analyze — Find the Cause (20%).....</b>                 | <b>8</b>  |
| Lab 3 — Analyze — Find the Root Cause.....                            | 8         |
| <b>Topic 05 — Improve — Fix the Cause (10%).....</b>                  | <b>9</b>  |
| Lab 4 — Improve — Choose and Pilot a Countermeasure.....              | 9         |
| <b>Topic 06 — Control — Hold the Gain (10%).....</b>                  | <b>10</b> |
| Lab 5 — Control — Hold the Gain and Hand Over.....                    | 10        |
| <b>Wrap-Up — The DMAIC Roadmap and Sustaining the Gain.....</b>       | <b>11</b> |
| <b>Next Steps.....</b>                                                | <b>12</b> |
| <b>Glossary.....</b>                                                  | <b>12</b> |

## Introduction

This Learner Guide accompanies the course Certified Lean Six Sigma White Belt (CLSSWB) Training (C142), conducted by Tertiary Infotech Academy Pte Ltd. This is a one-day Lean Six Sigma awareness course. It follows the DMAIC roadmap end to end — Define, Measure, Analyze, Improve, Control — with one hands-on lab in each of the five phases, and provides step-by-step instructions for every lab.

The course content is grounded in the body of knowledge published by The Council for Six Sigma Certification (CSSC) in 'Six Sigma: A Complete Step-by-Step Guide', so what you learn here matches the recognised White Belt standard. As a White Belt you are being prepared to CONTRIBUTE to an improvement team — to understand the language, recognise waste, and support the tools your team uses; the statistical analysis belongs to the Yellow, Green and Black Belt levels. Every lab uses one continuous scenario — the BrewBean Cafe morning rush, where customers queue up to 15 minutes and orders are sometimes made wrong. By the end of the day your five lab outputs form a complete improvement story: customer requirements and problem statement, process map and waste tally, root cause analysis, a selected countermeasure with a pilot plan, and a control plan.

## Course Learning Outcomes

- LO1: Describe the core concepts of quality, Lean, Six Sigma and Lean Six Sigma, and the belt roles in an improvement team.
- LO2: Explain the five phases of the DMAIC roadmap and what happens in each.
- LO3: Identify customer requirements and a clear problem statement in the Define phase.
- LO4: Describe how a process is mapped and measured, and identify the eight wastes in the Measure phase.
- LO5: Identify likely causes of a problem using 5 Whys and Fishbone analysis in the Analyze phase.
- LO6: Describe common improvement and control actions that fix a cause and hold the gain.

## Before You Start — Preparation

### What you need

- A laptop with a spreadsheet application (Microsoft Excel, Google Sheets or LibreOffice Calc), or simply pen and paper.
- A web browser for the interactive problem-solving tools listed in labs/tools.md.
- Printed or digital copies of this Learner Guide and the lab worksheets.
- A work process of your own to think about — the tools apply far better when the example is real.

## Verify your setup

No software installation is required for this course. Before you begin, confirm you can open a blank spreadsheet and reach the browser-based tools listed in labs/tools.md.

## Conventions used in every lab

- Every lab uses the same BrewBean Cafe scenario, so your outputs accumulate across the day.
- Each lab states what you will build; keep every output — later labs depend on earlier ones.
- Where a lab uses figures, the data is provided in the lab worksheet — no external data is needed.
- Use only process data you are authorised to share if you substitute your own workplace scenario.

## Topic 01 — Lean Six Sigma Foundations (20%)

Quality · Lean · Six Sigma · Lean Six Sigma · Belt roles · The DMAIC roadmap

### Key concepts

- What is Quality — Quality is meeting the customer's requirements — not just being free of defects.
- What is Lean — Lean removes waste so work flows faster to the customer.
- What is Six Sigma — Six Sigma reduces variation so results become consistent and predictable.
- Lean Six Sigma — The two combined: faster flow AND fewer defects, decided by data not opinion.
- Belt roles — White, Yellow, Green, Black and Master Black Belt — who does what on a team.
- The DMAIC roadmap — Define, Measure, Analyze, Improve, Control — the five-step improvement path.

## Topic 02 — Define — Understand the Problem (20%)

Voice of the Customer · CTQ · Problem statement · SMART goal · Project charter

### Key concepts

- Voice of the Customer — Ask customers what they need, and record it in their own words.
- Critical to Quality — Turn each customer need into something specific you can measure.
- Problem statement — What is wrong, where, since when, and how big — never the solution.

- SMART goal — Specific, Measurable, Achievable, Relevant and Time-bound.
- Project charter — A one-page summary of the problem, goal, scope and team.
- Scope — Agreeing what is in and out keeps a small project small.

## Lab 1 — Define — Customer Requirements and the Problem Statement

Learning outcome: Identify customer requirements and write a clear problem statement.

Goal: Start the BrewBean Cafe improvement the way every real project starts: find out what the customer actually wants, turn that into something measurable, and state the problem clearly enough that everyone agrees on it. The most common reason improvement projects fail is that the team never agreed what the problem was.

### What you'll build

A VOC-to-CTQ table, a written problem statement, a SMART goal and a simple scope table. (Tools: Voice of the Customer, CTQ, problem statement, SMART goal, project scope.)

### Step-by-step

1. Read the BrewBean Cafe scenario and underline every customer complaint you can find.
2. Write down five Voice of the Customer statements in the customer's own words (e.g. 'I'm late for work because the queue is too slow').
3. Turn each VOC statement into a measurable CTQ requirement — add a number and a unit (e.g. 'served within 5 minutes').
4. Write the problem statement using the template: what is wrong, where, since when, and how big. Do NOT write a solution.
5. Check your problem statement against the SMART test and rewrite the goal so it is Specific, Measurable, Achievable, Relevant and Time-bound.
6. Fill the in-scope / out-of-scope table so the team knows where the project stops.

### Test it

Every CTQ has a number and a unit, your problem statement contains no solution, and your goal has a metric, a baseline, a target and a date.

**Note:** Full commands and screenshots are in labs/lab-01-\*.md. Use only process data you are authorised to use if you substitute your own workplace scenario.

## Topic 03 — Measure — See What Is Really Happening (20%)

Process mapping · SIPOC · Types of data · Data collection · The eight wastes

## Key concepts

- Process mapping — Draw the steps as they really happen, not as the manual says.
- SIPOC — A one-page overview: Suppliers, Inputs, Process, Outputs, Customers.
- Types of data — Discrete data you count; continuous data you measure on a scale.
- Data collection — Agree what to record, who records it, and when — before you start.
- Check sheets — A simple tally form is the easiest reliable way to collect data.
- The eight wastes — DOWNTIME — Defects, Overproduction, Waiting, Non-utilised talent, Transport, Inventory, Motion, Extra-processing.

## Lab 2 — Measure — Map the Process and Spot the Waste

Learning outcome: Describe how the process runs and identify the eight wastes in it.

Goal: You cannot improve what you cannot see. Walk the BrewBean Cafe morning rush from the customer joining the queue to the drink being handed over, draw it as it really happens, and tally where time and effort are being lost. This is the phase that turns opinions about the problem into facts about the problem.

## What you'll build

A SIPOC overview, a step-by-step process map with timings, and a completed waste tally sheet. (Tools: SIPOC, process mapping, types of data, check sheets, the eight wastes (DOWNTIME).)

## Step-by-step

1. Agree the start and stop points of the process — start: customer joins the queue; stop: customer receives the drink.
2. Complete the SIPOC template: Suppliers, Inputs, Process (5-7 steps only), Outputs, Customers.
3. List the process steps in order and write the observed time next to each one.
4. Mark each step as value-adding (the customer would pay for it) or not value-adding.
5. Classify the data you collected: which figures are discrete (counted) and which are continuous (measured)?
6. Walk the process again with the DOWNTIME check sheet and tally every waste you observe against its type.
7. Identify which single waste type you tallied most often, and which process step it happens at.

## Test it

Your SIPOC has all five columns filled, every process step has a time, and every waste on your tally sheet is tagged to one of the eight DOWNTIME types.

**Note:** Full commands and screenshots are in labs/lab-02-\*.md. Use only process data you are authorised to use if you substitute your own workplace scenario.

## Topic 04 — Analyze — Find the Cause (20%)

Root cause · 5 Whys · Fishbone diagram · Pareto chart · Variation

### Key concepts

- Symptom vs cause — Treating the symptom makes the problem come back; treating the cause does not.
- Root cause — The cause that, once removed, stops the problem recurring.
- 5 Whys — Keep asking 'why?' until you reach something you can actually act on.
- Fishbone diagram — Sort possible causes into Manpower, Method, Machine, Material and Measurement.
- Pareto chart — Roughly 80% of the problem usually comes from 20% of the causes.
- Variation — Some variation is normal in the process; some has a specific assignable cause.

### Lab 3 — Analyze — Find the Root Cause

Learning outcome: Identify likely causes using 5 Whys and Fishbone analysis.

Goal: You now know WHERE the delay happens. This lab asks WHY it happens. Most teams jump straight from a symptom to a solution and end up fixing the wrong thing — the queue is long, so hire more staff — when the real cause was something else entirely. Here you drill past the symptom to a cause you can actually act on.

### What you'll build

A completed 5 Whys chain, a Fishbone diagram with causes sorted by category, and a shortlist of likely root causes. (Tools: Symptom vs cause, 5 Whys, Fishbone (5M) diagram, Pareto chart, common vs special cause variation.)

### Step-by-step

1. Write the symptom from your Lab 2 findings at the top of the 5 Whys template (e.g. 'customers wait 15 minutes').
2. Ask 'why does that happen?' five times, each answer becoming the next question. Stop when you reach something you can act on.
3. Draw the Fishbone diagram and write the problem in the head of the fish.
4. Brainstorm possible causes onto the five bones: Manpower, Method, Machine, Material and Measurement.

5. Read the supplied Pareto chart of BrewBean complaint types and state which few causes account for most of the problem.
6. Decide for each shortlisted cause whether it is common cause (happens every day) or special cause (a one-off event).
7. Shortlist the two or three causes best supported by the data you collected in Lab 2.

### Test it

Your 5 Whys chain ends in something the team can actually act on, every Fishbone cause sits under one of the five categories, and each shortlisted cause is backed by an observation, not an opinion.

**Note:** Full commands and screenshots are in labs/lab-03-\*.md. Use only process data you are authorised to use if you substitute your own workplace scenario.

## Topic 05 — Improve — Fix the Cause (10%)

Generating solutions · 5S · Mistake proofing · Standard work · Piloting

### Key concepts

- Generating solutions — Brainstorm against the proven cause — never against the symptom.
- Choosing a solution — Compare ideas on impact, effort and risk before committing.
- 5S — Sort, Set in order, Shine, Standardise, Sustain — a tidy workplace for physical or digital work.
- Mistake proofing — Poka-Yoke makes the error hard or impossible to make.
- Standard work — Write the better method down so everyone does it the same way.
- Piloting — Try the change small first, so mistakes stay cheap.

### Lab 4 — Improve — Choose and Pilot a Countermeasure

Learning outcome: Describe improvement actions that address the identified cause.

Goal: A cause is only worth finding if something changes because of it. Take the root causes you shortlisted in Lab 3, generate countermeasures against them, and pick one that is small enough to try next week. The discipline here is to solve the CAUSE you proved, not the symptom you started with.

### What you'll build

A brainstormed countermeasure list, an impact/effort grid, one selected countermeasure and a simple pilot plan. (Tools: Brainstorming, impact/effort screening, 5S, mistake proofing (Poka-Yoke), standard work, piloting.)

## Step-by-step

1. Write your top root cause from Lab 3 at the top of the sheet — every idea must address THIS cause.
2. Brainstorm at least six countermeasures without judging any of them yet.
3. Check whether 5S would help: is anything at the counter hard to find, out of place or untidy?
4. Check whether mistake proofing would help: could the wrong order be made impossible rather than merely discouraged?
5. Plot every idea on the impact/effort grid and pick one from the high-impact, low-effort corner.
6. Write the improved method as standard work — short numbered steps anyone on shift could follow.
7. Write a one-week pilot plan: what changes, who runs it, what you will measure, and how you will know it worked.

## Test it

Your selected countermeasure clearly addresses the root cause from Lab 3 (not the symptom), and your pilot plan names a measure, an owner and a review date.

**Note:** Full commands and screenshots are in labs/lab-04-\*.md. Use only process data you are authorised to use if you substitute your own workplace scenario.

## Topic 06 — Control — Hold the Gain (10%)

Control plan · Visual management · SOPs · Team huddles · Handover

### Key concepts

- Why Control matters — Without Control, processes quietly drift back to the old way.
- Control plan — What we measure, the target, how often, who owns it, and what to do if it slips.
- Visual management — Make performance visible so problems are noticed the same day.
- Standard operating procedures — Written instructions that lock in the improved method.
- Team huddles — Short regular stand-ups catch problems while they are still small.
- Handover — Give the improved process back to the people who run it every day.

## Lab 5 — Control — Hold the Gain and Hand Over

Learning outcome: Describe the control actions that sustain an improvement.

Goal: Most improvements work for a fortnight and then quietly disappear. Control is what stops that happening. Close the BrewBean Cafe story by deciding what will be measured from now on, who owns it, what happens when it slips, and how the improved process is handed back to the people who run it every day.

### What you'll build

A completed control plan, a visual management board sketch, and a one-page summary of the whole DMAIC story. (Tools: Control plan, visual management, SOPs, team huddles, handover, one-page summary.)

### Step-by-step

1. Choose the ONE measure that tells you the improvement is still working (e.g. average wait time at 8:00am).
2. Complete the control plan template: measure, target, how often it is checked, and who owns it.
3. Write the reaction plan — the exact steps to take when the measure misses target.
4. Sketch a simple visual board that would make this measure visible to the whole shift.
5. Turn your Lab 4 standard work into a short SOP the cafe could actually pin up.
6. Plan a five-minute daily huddle: who attends, what is reviewed and how long it lasts.
7. Summarise the whole project on one page — problem, cause, countermeasure, result and control — and name who you hand it over to.

### Test it

Your control plan names a measure, a target, a frequency and an owner; your reaction plan says what to DO when it slips; and your one-page summary tells the complete DMAIC story from problem to handover.

**Note:** Full commands and screenshots are in labs/lab-05-\*.md. Use only process data you are authorised to use if you substitute your own workplace scenario.

## Wrap-Up — The DMAIC Roadmap and Sustaining the Gain

These cross-cutting themes run through every phase of the course. Study this section alongside the labs so the same approach transfers from the BrewBean Cafe scenario to a real improvement in your own workplace.

### The DMAIC roadmap at White Belt depth

- Define — capture the Voice of the Customer, translate it into measurable CTQs, and agree a problem statement with no solution in it.
- Measure — map the process as it really happens (SIPOC, process map) and tally the eight wastes with a check sheet.
- Analyze — drill past the symptom with 5 Whys and a Fishbone, and read a Pareto chart to see the vital few causes.
- Improve — brainstorm against the proven cause, screen ideas on impact vs effort, and pilot the change small first.
- Control — lock in the gain with a control plan, visual management, SOPs, team huddles and a proper handover.

### Habits that make improvement stick

- Measure before you change — without a baseline you cannot prove the improvement worked.
- Attack the process, not the people — the system produces the result you are seeing.
- Never write a solution into the problem statement.
- Ask 'why?' past the symptom until you reach something the team can act on.
- Standardise the improved method, or the process will drift back within weeks.

### Next Steps

- First pass: complete every lab, following the steps in each lab worksheet.
- Second pass: repeat the five phases on a small process from your own workplace.
- Keep your improvement story together — the VOC/CTQ table, problem statement, process map, waste tally, root cause, pilot plan and control plan.
- Progress towards Yellow Belt for deeper data collection, process metrics and analysis tools.

### Glossary

- **Quality** — Meeting the customer's requirements — not just being free of defects.
- **Lean** — A method to maximise customer value by systematically removing waste and improving flow.
- **Six Sigma** — A data-driven method to reduce variation so results become consistent and predictable.
- **DMAIC** — Define, Measure, Analyze, Improve, Control — the five-step Lean Six Sigma improvement roadmap.
- **Belt roles** — White, Yellow, Green, Black and Master Black Belt — increasing levels of Lean Six Sigma capability.
- **VOC** — Voice of the Customer — customer needs captured in the customer's own words.

- **CTQ** — Critical to Quality — a specific, measurable requirement translated from a VOC need.
- **Problem statement** — What is wrong, where, since when, and how big — never the solution.
- **SMART goal** — Specific, Measurable, Achievable, Relevant and Time-bound.
- **Project charter** — A one-page summary of the problem, goal, scope and team.
- **SIPOC** — A one-page process overview: Suppliers, Inputs, Process, Outputs, Customers.
- **Check sheet** — A simple tally form — the easiest reliable way to collect process data.
- **DOWNTIME** — The eight wastes: Defects, Overproduction, Waiting, Non-utilised talent, Transport, Inventory, Motion, Extra-processing.
- **Value-added** — An activity the customer would be willing to pay for, done right the first time.
- **Root cause** — The cause that, once removed, stops the problem recurring.
- **5 Whys** — Asking why repeatedly to drill from a symptom down to an actionable root cause.
- **Fishbone diagram** — A cause-and-effect diagram sorting causes into Manpower, Method, Machine, Material and Measurement.
- **Pareto chart** — A bar chart showing that roughly 80% of the problem comes from 20% of the causes.
- **Variation** — The spread in process output; common cause is inherent, special cause is assignable.
- **5S** — Sort, Set in order, Shine, Standardise, Sustain — workplace organisation for physical and digital work.
- **Poka-Yoke** — Mistake proofing — making an error difficult or impossible to make.
- **Standard work** — The documented best-known method so anyone can repeat it the same way.
- **Piloting** — Trying a change small first, so mistakes stay cheap.
- **Control plan** — Metric, target, monitoring method, frequency, owner and reaction plan.
- **Visual management** — Making performance visible so problems are noticed the same day.